

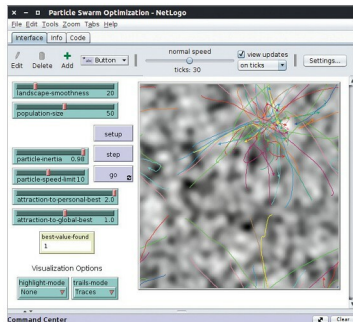


Figure 2: NetLogo interface 2

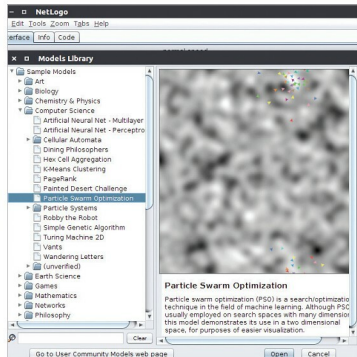


Figure 3: NetLogo interface 3

from secondary school students to graduates can make use of this software to study their field of interest.

Installation

Installation of NetLogo in Ubuntu 14.04 is easy. First, download the NetLogo package from <http://ccl.northwestern.edu/netlogo/>. I have a 64-bit Linux computer, so I have downloaded the package accordingly. For NetLogo version 6.0.1, memory size of 197 MB is required. The downloaded file is compressed and must be extracted in the home directory (preferably in the *opt* folder). Once extracted, you are ready to use NetLogo on your computer. In the NetLogo directory, double-click on the NetLogo executable to open the GUI of the software.

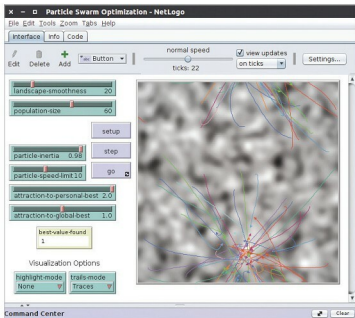


Figure 4: NetLogo interface 4

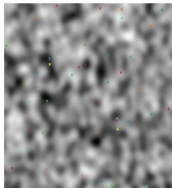


Figure 5: Simulation results 1

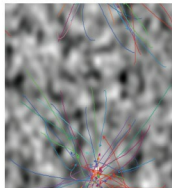


Figure 6: Simulation results 2

Particle swarm optimisation (PSO) in NetLogo

PSO is a nature inspired swarming theory based algorithm invented by Kennedy and Eberhart in the 1990s. Flocks of birds migrating from place to place in search of food and shelter are the basis for this algorithm. Fortunately, NetLogo has inbuilt model libraries where many models like PSO are readily available. The model demonstrates how the true optimum value 1 is found for various parameter settings like landscape smoothness (structure of valleys and hills), population size (number of birds), etc.

The NetLogo interface is shown in Figure 1. Figure 3 shows the PSO model file available in its library. Figure 2 shows the PSO finding its true value in 30 ticks. Figure 4 shows the same attained in 22 ticks. Note that fewer ticks are observed for a population of 10. Figures 5 and 6 show the above-mentioned initial and final results. **END** 

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